Asad Razzaq

asad.razzaq@kuas.ac.jp | ORCID | LinkedIn | Google Scholar | Kyoto, Japan
Nationality: Pakistani | **Date of Birth: 15/05/1996**

Research Interests

Computational Pathology · Medical Image Analysis · Multimodal Data Fusion (Histology, Genomics, Clinical Data) · Foundation Models for Medical AI · Cancer Diagnosis, Prognosis & Biomarker Discovery · AI-Assisted Clinical Decision Support · Medical Robotics

Education

University of Toyama — **Toyama, Japan**
Ph.D. in Engineering, Department of Mechanical and Intellectual Engineering — *Oct 2022 – Sept 2025*
Advisor: Prof. Toshiyuki Yasuda
Thesis: Evolutionary Design of Collective Behavior for Heterogeneous Swarm Robotic Systems.
Developed AI-driven frameworks for multi-agent robotic systems, enabling autonomous coordination, distributed exploration, and adaptive control. Utilized neural networks and optimization techniques to learn scalable policies for complex environments, with relevance to robotic navigation, physical AI, and real-world deployment.

University of Toyama — **Toyama, Japan**
Research Student in Control Systems Engineering Laboratory — *April 2022 – Oct 2022*

Government College University Faisalabad — **Faisalabad, Pakistan**
M.S. in Computer Science — *Aug 2017 – Mar 2020*
Thesis: Impacts of Blockchain in Governance with Respect to Finance

Government College University Faisalabad — **Faisalabad, Pakistan**
B.S. in Computer Science — *Oct 2013 – Aug 2017*

Iqra College Sadiqabad — **Sadiqabad, Pakistan**
F.Sc. Pre-Engineering — *Aug 2011 – May 2013*

Bismillah Higher Secondary School — **Sadiqabad, Pakistan**
Matriculation (Science) — *Mar 2009 – Mar 2011*

Research & Industry Experience

Kyoto University of Advanced Science (KUAS) — **Kyoto, Japan**
Specially-Appointed Researcher, Department of Mechanical and Electrical Systems — *Dec 2025 – current*
Areas of Research NISAR LAB: Medical Robotics, AI-powered Human-Machine Systems, Haptics, Soft Manipulators/Actuators

Highlight: Exploring applications of AI and machine learning in multi-robot systems, autonomous exploration, and real-world environmental robotics.

University of Toyama — **Toyama, Japan**
Research Assistant — *Oct 2022 – Sept 2025*
- Led research on evolutionary design of collective behavior in heterogeneous swarm robotic systems.
- Developed and optimized neural controllers for tasks such as cooperative transport and collective exploration.

Souya Corporation — **Ishikawa, Japan**
AI Systems Engineering Intern — *Sept 2025*
- Assisted in the development of AI-driven software solutions integrating deep learning and data analytics pipelines for enterprise clients.
- Collaborated with cross-functional teams spanning AI, software development, and system integration.
- *Highlight: AI System Development, Deep Learning, Data Science, PoC Design, Python, Software Integration.*

IRIDIA Lab, Université Libre de Bruxelles — **Brussels, Belgium**
Visiting Ph.D. Student — *Mar 2025*
- Collaborated on a focused research project under the supervision of Professor MARCO DORIGO, a pioneer in swarm intelligence. Explored the integration of blockchain technologies in swarm robotics to enhance decentralized coordination, trust, and data integrity

Approsoft Web Development and Design Company — **Faisalabad, Pakistan**
IT Executive — *Apr 2020 – Mar 2022*
- Managed full-cycle web development projects and delivered tailored client software solutions.
- Improved communication workflows between technical and non-technical teams, enabling efficient project execution.
- *Highlight: Web Development, Project Management, Client Communication, Software Deployment.*

Publications

Journal Papers

[J1] Razzaq. A, T. Yasuda, "Evolutionary design of group organization and controllers in heterogeneous robotic swarms," *Artificial Life and Robotics*, vol. 31, pp. 294–301, 2026.
DOI: 10.1007/s10015-025-01083-5 [Paper]

[J2] Asad. R, T. Hayakawa, T. Yasuda, "Evolutionary Design of Cooperative Transport Behavior for a Heterogeneous Robotic Swarm," *Journal of Robotics and Mechatronics*, vol. 35, pp. 1007–1015, 2023.
DOI: 10.20965/jrm.2023.p1007 [Paper]

[J3] Razzaq. A, M. M. Khan, R. Talib, A. D. Butt, N. Hanif, S. Afzal, M. R. Raouf, "Use of Blockchain in Governance: A Systematic Review of the Literature," *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 5, 2019.
DOI: 10.14569/IJACSA.2019.0100585 [Paper]

International Conferences (Peer reviewed)

[C1] Asad Razzaq, Toshiyuki Yasuda, et al., "Knowledge Distillation for Controller Versatility in Robotic Swarms," *ANTS26*, Germany, 2026.

[C2] Asad Razzaq, Senshu Wakayama, et al., "Evolutionary Approach for Optimization of Perceptual Abilities and Group Composition in Heterogeneous Robotic Swarms," *SWARM*, Germany, Sept 2025.

[C3] Asad Razzaq, Toshiyuki Yasuda,"Investigating Swarm Behavior through Group Compositions with Perceptual Variability in Evolutionary Swarm Robotics System," *SICE FES*, Thailand, Sept 2025. [Paper]

[C4] Asad Razzaq, Toshiyuki Yasuda, "Evolutionary Automatic Design of Heterogeneous Robotic Swarm Composition for Cooperative Transport Behavior," *SWARM 2024*, Kyoto, Japan, Sept 2024. [Paper]

[C5] Asad Razzaq, Toshiyuki Yasuda, "Neural Controller Optimization for Evolutionary Heterogeneous Swarm Exploration," *SICE FES 2024*, Kochi, Japan, Aug 2024. [Paper]

[C6] Asad Razzaq, Hayakawa Tomohiro, Toshiyuki Yasuda, "Investigating The Influence of Individual Robot Perception Capability on Swarm Behavior in an Evolutionary Swarm Robotic System," *Proceedings of AROB-ISBC-SWARM 2024*, Oita Beppu, Japan, January 2024. [Paper]

[C7] Asad Razzaq, Takizawa Minato, et al., "Evolutionary Acquisition of Collective Exploration Behavior in Heterogeneous SRS: Investigation on Neural Controller Configuration," *Proceedings of AROB-ISBC-SWARM 2023*, pp. 1424–1429, Oita, Japan, Jan 2023. [Paper]

[C8] Muhammad Azhar Mushtaq, Abid Sultan, Asad Razzaq, Muhammad Inaam-ur-Rehman, "New Cryptographic Algorithm Based on CFA," *Proceedings of the 2019 4th International Conference on Big Data and Computing*, Beijing, China, May 2019. [Paper]

Poster

[P1] Asad Razzaq, Toshiyuki Yasuda, "Evolutionary Design of Artificial Swarms as Sustainable Systems," *Toyama Academic GALA 2024 University of Toyama*, Toyama, Japan. [Poster]

Thesis

[T1] Asad Razzaq. **Evolutionary Design of Collective Behavior for Heterogeneous Swarm Robotic Systems,** Ph.D. Dissertation, Department of Mechanical and Intellectual Engineering, University of Toyama, July 2025.

[T2] Asad Razzaq. **Impacts of Blockchain in Governance with Respect to Finance,** MS Thesis, Department of Computer Science, Government College University Faisalabad, Pakistan, Mar 2020.

[T3] Asad Razzaq. **Urdu Steamer,** BS Project, Department of Computer Science, Government College University Faisalabad, Pakistan, July 2017.

Awards & Recognition

Toyama University Next Generation Researchers Challenging Research Program: Funding support awarded by JST SPRING for graduate students engaging in challenging and interdisciplinary research, Grant Number JPMJSP2145 (Apr 2023 – Sept 2025).

Outstanding Student Award: Presented by the Measurement and Automatic Control Society, Hokuriku Branch, Japan, for excellence in academic performance and research achievements, (2024).

PhD Vice Valedictorian: Awarded by the University of Toyama to the second-highest-ranking doctoral graduate in recognition of outstanding academic excellence and scholarly contributions (Sept 2025).

SWARM 2025 Travel Grant: Selected for a travel grant to attend the SWARM 2025 conference (Sept 2025).

Prime Minister's Laptop Scheme: Awarded a laptop under the Government of Pakistan's merit-based initiative to support high-achieving university students across the region (2019).

Technical Skills

Languages: English (Proficient C1), Japanese (Basic N5), Urdu (Native)
Programming: C/C++, C#, Python, MATLAB, JavaScript, LaTeX
Software: Systems (Linux, Windows, V-Rep/ROS/ROS2,Gazebo, Isaac Sim, Isaac Lab, Genesis, MuJoCo, Drake), MONAI, OpenSlide, Pytorch, Tensorflow, CUDA, OpenCV, Git, Visual Studio, Unity, Gazebo, Eagle, Altium Designer, Mission Planner, Ardupilot
Medical AI Tools: PyTorch, MONAI, OpenSlide, CUDA, OpenCV
Hardware: Raspberry Pi, Arduino, Beaglebone, Udoo, Pixhawk, Ardupilot, DJI Mavic, Parrot Bebop
Current Focus: Exploring deep learning and multimodal AI approaches, including medical image analysis, representation learning, and data-driven modeling for real-world applications.

Volunteering

- Gave a talk about career, graduate study, and scholarship seminars for Pakistani student communities. 2022 - present.

References

Prof. Toshiyuki Yasuda
University of Toyama, Japan
Ph.D. Supervisor
Email: yasuda@eng.u-toyama.ac.jp
Web: Research Profile

Assoc. Prof. Sajid Nisar
Kyoto University of Advanced Science (KUAS), Japan
Postdoctoral Supervisor
Email: sajid.nisar@kuas.ac.jp
Web: Faculty Profile

Dr. Allah Nawaz
Joslin Diabetes Center, Harvard
School Teacher
Email: allah.nawaz@joslin.harvard.edu
Web: Faculty Profile